Xi'an Jiaotong-Liverpool University 西交利物浦大学

MEC202 GROUP PROJECT -- Group 34

Indoor and outdoor Pedestrian Positioning

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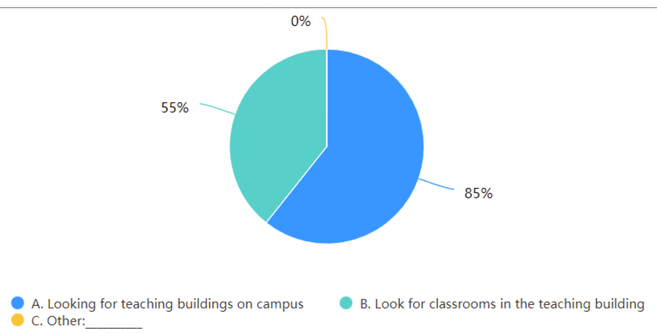
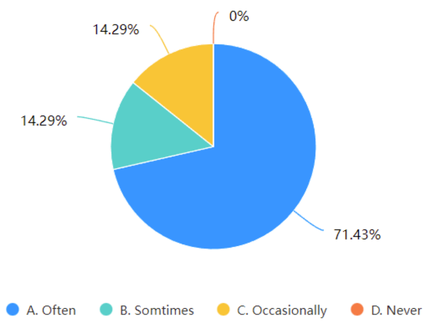
Introduction

Indoor Positioning: A Smartphone Solution
Navigating indoor spaces can be challenging without GPS. Our project offers a user-friendly smartphone app using Pedestrian Dead Reckoning (PDR) technology for accurate indoor positioning. Discover seamless navigation with our innovative solution, enhancing user experience in diverse environments.

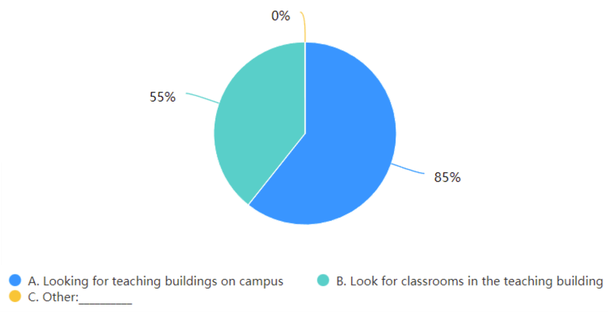
User survey:

Has the online map ever been unavailable due to poor signal, especially in the building?





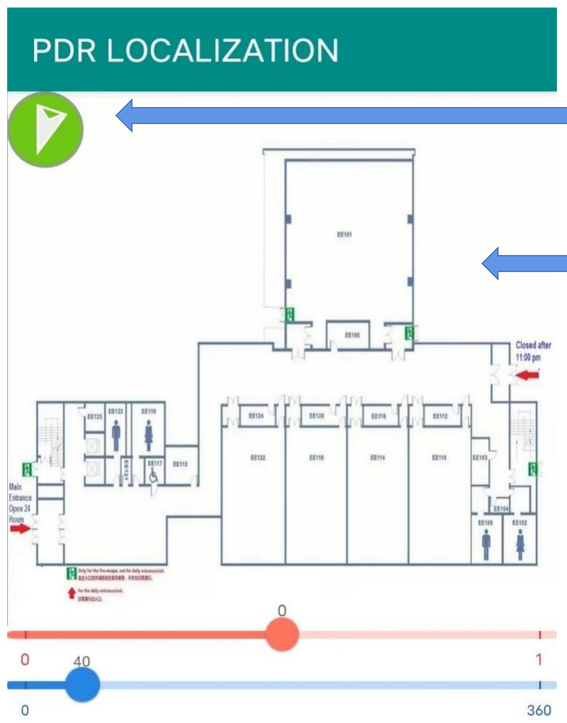
How often do you use online maps, such as Amap?



In which of the following scenarios would you use an online map in your school?(multiple options can be selected)

Product Design and Function

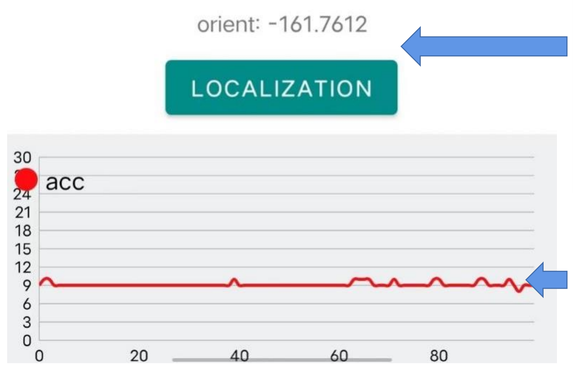
product design drawing



Compass, based on calling magnetometer data

An indoor map used to display the user's current location, in the demo we use the map on the first floor of the EE building

Two axes that can be manually adjusted, corresponding to the step value and direction from top to bottom, are used in addition to the algorithm and sensor data. Users can manually adjust the two values to obtain more accurate data to achieve positioning.



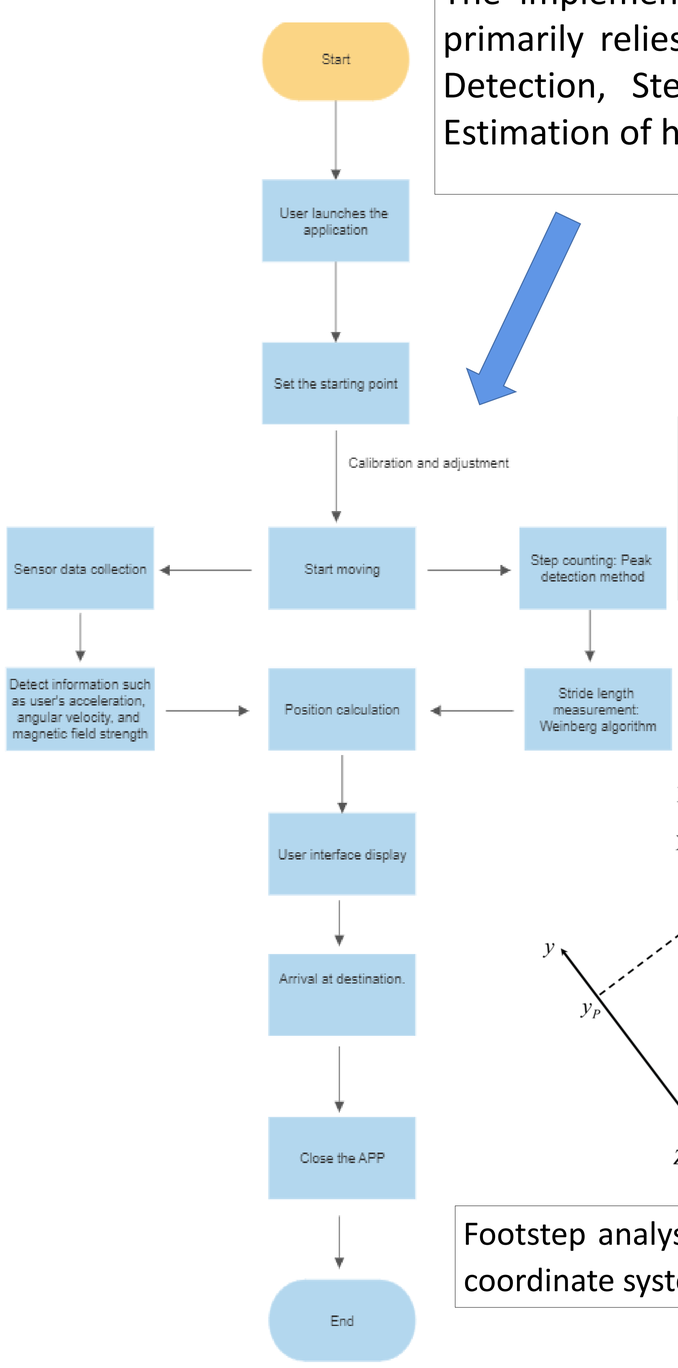
Reset the location by clicking the button, and then the user needs to manually click on his location on the map to obtain the initial location for further positioning.

Graph showing step size visually, step change recorded

Program Logic and Algorithms

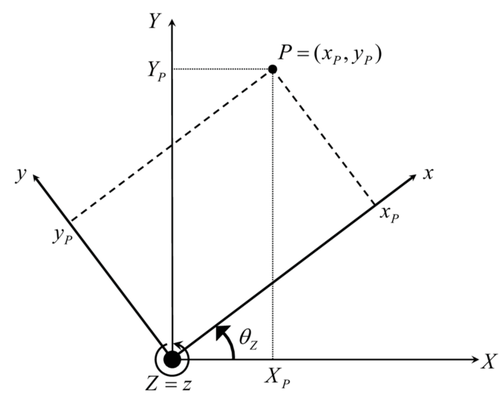
Logic design

The implementation of PDR functionality primarily relies on three algorithms: Step Detection, Step Length Estimation, and Estimation of heading.



Weinberg Method: k is a scaling factor, Amax is the maximum acceleration value, and Amin is the minimum acceleration value

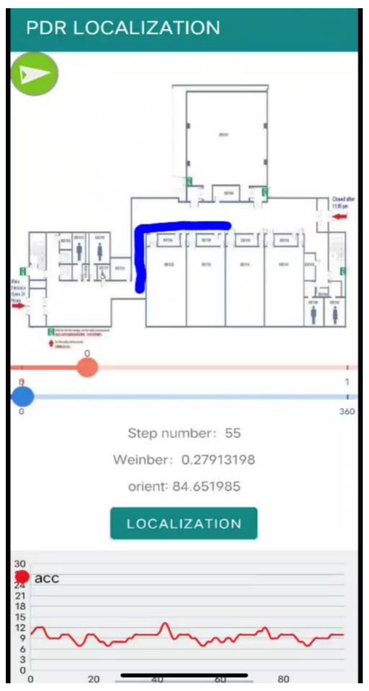
$$Step_size = k\sqrt[4]{A_{\max} - A_{\min}}$$



Footstep analysis and acceleration in coordinate systems

Prototype

Prototype offline demo



The usability of the prototype was confirmed through the offline test in the corridor of the EE building. The blue line in the figure proves that the sensor is correctly invoked and the correct determination direction at the corner is achieved.